SCHOOL OF ENGINEERING & COMPUTING

THERMAL PERFORMANCE OF METAL-PRINTED HYBRID GYROID-DIAMOND TPMS STRUCTURES EMBEDDED WITH PCM

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BACKGROUND

- Phase change materials (PCMs) are widely studied for thermal energy storage due to their high latent heat capacity, which allows them to absorb and release large quantities of energy near-isothermally during melting and solidification.
- Most paraffin-based PCMs suffer from inherently low thermal conductivity (typically 0.2–0.3 W/m·K)
- Triply Periodic Minimal Surface (TPMS) lattice structures have emerged as a promising class of heat transfer enhancers due to their high surface-area-to-volume ratios, smooth curvature, and inherent connectivity.
- Two TPMS geometries of particular interest are the Gyroid and the Diamond surface as shown in Fig. 1.
- Both offer favorable thermal characteristics, they differ in topology, pore connectivity, and surface orientation
- Suggesting that combining them in a hybrid configuration could exploit complementary heat transfer pathways within a single structure.

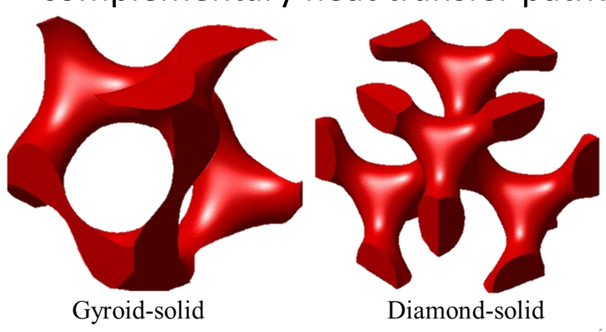

Figure 1. TPMS Unit Cell

OBJECTIVES

- Design a hybrid Gyroid-Diamond TPMS geometry that smoothly alternates between the two topologies along a curved spatial path, using implicit surface modeling in MATLAB.
- Fabricate samples at three relative densities (25%, 35%, 45%) via metal paste deposition (MPD) metal additive manufacturing to isolate the effect of density on thermal performance.
- Characterize PCM melting behavior within each sample under controlled heating conditions using embedded thermocouples.
- Evaluate how relative density governs phase change progression and thermal response across the hybrid TPMS geometry.

TPMS GEOMETRY DESIGN

- Hybrid structure alternates between gyroid/diamond across six discrete bands arranged along a curved S-shaped path through the sample volume as shown in Fig. 3.
- A binary step transition was used at each band boundary, which produced cleaner, more printable interfaces than continuous blending approaches.
- The S-shaped path was parameterized explicitly, with band membership assigned to each voxel based on its projection along the path's arc length. The finalized geometry was exported as STL files and combined with an outer shell offset in PrusaSlicer prior to fabrication.

FABRICATION AND EXPERIEMNTAL SETUP

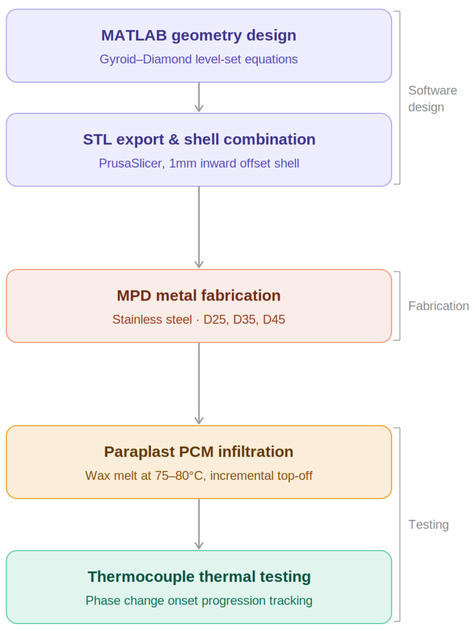

Figure 2. Process plan for design and experimentation

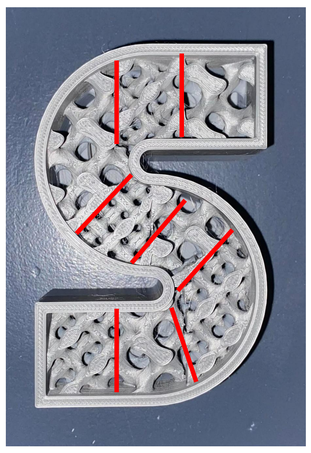

Figure 3. 3D printed hybrid TPMS structure

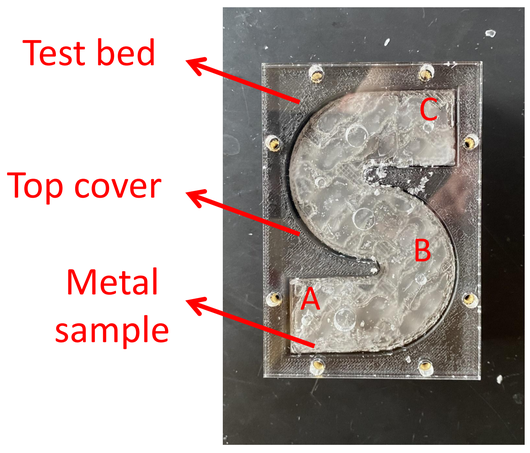

Figure 4. Metal sample infiltrated with PCM for testing

EXPECTED OUTCOMES & FUTURE WORK

- Results will be evaluated by tracking phase change onset progression, specifically, the sequence and timing with which melting is detected at the inlet (A), midpoint (B), and outlet (C) thermocouples, as a measure of how effectively each sample distributes heat through the PCM volume.
- It is expected that higher relative density samples (45%) will demonstrate faster phase change propagation due to increased solid conduction pathways, while lower density samples (25%) may show more localized melting near the heat source.
- This work contributes to a broader design framework for geometry-driven thermal management components, with implications for battery thermal regulation, electronics cooling, and portable energy storage systems.

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